$$\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z}), \qquad \mathbb{Z}_N = \mathbb{Z}/N\mathbb{Z}, \qquad h = 2\pi/N$$

$$\begin{tikzcd}[column sep=huge, row sep=huge]
\text{Continuous, nonperiodic} \arrow[r, "\text{Periodize}", "\text{period } 2\pi"'] \arrow[d, "\text{Sample}", "\text{step } h"'] & \text{Continuous, periodic} \arrow[d, "\text{Sample}", "\text{step } h"'] \\
\text{Discrete, nonperiodic} \arrow[r, "\text{Periodize}", "\text{period } N"'] & \text{Discrete, periodic}
\end{tikzcd}$$

Continuous, nonperiodic

$$f(x) \overset{\mathcal{F}}{\longleftrightarrow} \widehat{f}(\omega), \qquad x \in \mathbb{R}, \quad \omega \in \mathbb{R}$$

Continuous, periodic

$$p(x) \overset{\text{Fourier series}}{\longleftrightarrow} c_k, \qquad x \in \mathbb{T}, \quad k \in \mathbb{Z}$$

Discrete, nonperiodic

$$a_n \overset{\text{Fourier series}}{\longleftrightarrow} A(\theta), \qquad n \in \mathbb{Z}, \quad \theta \in \mathbb{T}$$

Discrete, periodic

$$b_n \overset{\text{DFT}}{\longleftrightarrow} B_k, \qquad n \in \mathbb{Z}_N, \quad k \in \mathbb{Z}_N$$

$$p(x) = \sum_{j\in\mathbb{Z}} f(x + 2\pi j), \qquad a_n = f(hn), \qquad b_n = p(hn) = \sum_{j\in\mathbb{Z}} a_{n+jN}$$

Periodization in one domain corresponds to sampling in the other

$$c_k = \frac{\widehat{f}(k)}{2\pi}, \qquad A(\theta) = \sum_{n\in\mathbb{Z}} a_n e^{-in\theta} = \frac{1}{h} \sum_{j\in\mathbb{Z}} \widehat{f}\left(\frac{\theta + 2\pi j}{h}\right)$$

$$B_k = \sum_{n=0}^{N-1} b_n e^{-2\pi i n k/N} = A\left(\frac{2\pi k}{N}\right) = N \sum_{j\in\mathbb{Z}} c_{k+jN}$$

The displayed identities hold, for example, for a rapidly decaying smooth signal $f \in \mathcal{S}(\mathbb{R})$.